QUANTEINSTEIN ENTERPRISE OS V3.0

INSTITUTIONAL EXECUTION SUBSTRATE & PRE-TRADE RISK CORE

Quanteinstein Technologies, Inc.
Architecture Brief & Due Diligence
Release: Production C++20 Core

Institutional Executive Summary: Quanteinstein is a high-performance C++20 execution engine and private single-tenant deployment substrate engineered for emerging quantitative hedge funds ($2M–$50M AUM), crypto prop desks, and systematic market makers. It bypasses 9–14 months of internal systems engineering by providing cacheline-partitioned shared-memory IPC, live zero-allocation crypto/FX ingestion, and native FIX 4.4 DMA connectivity.

INGESTION THROUGHPUT	STEADY-STATE MEMORY	LIVE WIRE-TO-RING	CHANNEL CAPACITY
2.71M / sec	**0 Bytes**	**1.50 µs**	**65,536**
Sustained 5M real burst	Strict zero-alloc assertion	p50 live Binance & FX ticks	24MB pre-alloc per shard

1. DETERMINISTIC C++20 CORE SUBSTRATE

- **Lock-Free POSIX SHM:** Branchless bitwise index masking (`head & (CAPACITY - 1)`) across pre-allocated `/dev/shm` ring channels.
- **Cacheline-Partitioned Memory:** 384-byte `alignas(64)` structs: `write_head` (CL#1, 64B), `read_tail` (CL#2, 128B), and handshake flags (CL#5, 320B). Eliminates cross-core false sharing.
- **Multi-Channel Symbol Sharding:** Branchless 64-bit FNV-1a symbol hashing routes flow across 4 independent SPSC rings bound to isolated CPU cores.
- **Kernel-Bypass Socket Layer:** Zero-jitter TCP configuration (`TCP_NODELAY`, 4MB buffers, Linux `SO_BUSY_POLL = 50µs`) with native thread affinity pinning.

2. REGULATORY CONTROLS & INGESTION LAYER

- **Live WebSocket Ingestion:** Zero-allocation in-place string scanner for Binance Spot & Futures feeds (`@aggTrade` & `@depth20`).
- **Institutional Forex Streaming:** Live major currency pair ticks (EUR/USD, GBP/USD, USD/JPY, USD/CHF) with microsecond fixed-point pip scaling.
- **SEC 15c3-5 & Reg SHO:** Real-time pre-trade risk filters (capital collars, size limits) and dynamic cryptographic locate decrement ledger.
- **Deterministic Fault Recovery:** Write-Ahead Journaling (WAL) with automatic sequence gap detection, FIX Tag 35=2 Resend, and Tag 35=4 GapFill.

VERIFIED OPERATIONAL DUE DILIGENCE (ODD) TELEMETRY

BENCHMARKED SUBSTRATE LAYER	INGESTION THROUGHPUT	HARDWARE LATENCY PROFILE	MEMORY & CONCURRENCY AUDIT
C++20 SPSC Ingestion Core (8 Producer Threads, 10,000 Symbols)	`2,708,529 orders/sec`	Min: 19 ns | p50: 363 ns p90: 1.47 µs | p99: 4.78 µs	0 bytes heap allocated 0 bus lock / split conflicts
Live Binance WSS & Forex Bridge (BTC, ETH, SOL, EUR/USD, GBP/USD)	`442 msgs/sec` (Live exchange pacing)	Min: 139.6 ns | p50: 1.50 µs p90: 10.5 µs | p99: 26.5 µs	0 dropped packets (0.000%) 0 backpressure retries
Chaos Ingestion & Poison Fuzzing (NaN, INF, Corrupted Magic, Short LOC)	`100,000 toxic payloads` burst-injected	Catch Rate: 100.0% 0 core dumps / 0 crashes	100% WAL crash recovery Tag 35=2 Resend verified

DESIGN PARTNER COMMERCIAL SOURCE LICENSE AGREEMENT (CSLA)

- **Full C++20 Source Access:** Complete source code for MultiChannelRouter, ShmChannel, FixEngine, and SocketConfig.
- **Single-Tenant Private VPC:** Deploys inside the fund's own infrastructure (zero multi-tenant alpha leakage).
- **Engineering Support:** 15 hours dedicated architectural integration and onboarding support.
- **Internal Enterprise Rights:** Non-exclusive perpetual internal runtime rights with 1 year of continuous core updates.

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